

Mandazi: Atomic Data Settlement via PLONKish-Proved ElGamal over Interledger

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Abstract

Zero-Knowledge Contingent Payment (ZKCP) protocols, originally introduced by Maxwell [1], enable the fair exchange of digital assets and financial liquidity without relying on a trusted third party. However, deploying general-purpose ZKCP mechanisms over modern Open Payments routing architectures introduces a fundamental performance mismatch: high-throughput payment settlement networks dictate the use of signature schemes over non-native fields (e.g., Ed25519 over $\mathbb{F}_{2^{255}-19}$ via the G NAP protocol [6]), whereas efficient zero-knowledge proving systems typically utilize distinct scalar fields (e.g., BN254 or BLS12-381) to minimize constraint density. In this paper, we introduce **Mandazi**, an atomic, dual-curve cryptographic framework that unlocks private, per-query data settlement directly over existing Interledger Protocol (ILP) [5] and Rafiki routing structures. Mandazi resolves the cross-curve impedance bottleneck by implementing a lane-isolated architecture. The underlying protocol isolates the Ed25519 financial routing layer out-of-circuit, while simultaneously binding the data validation and delivery mechanics within a monolithic circuit. Our arithmetization utilizes a PLONKish system [4] optimized via custom gates and lookup tables to verify a Poseidon-based Merkle membership proof of an authoritative data registry leaf. Concurrently, the circuit enforces an integrated ElGamal-style asymmetric key exchange over an embedded Baby Jubjub twisted Edwards curve, producing a deterministic pseudo-random keystream that encrypts the target data payload. Crucially, the circuit architecture imposes absolute witness cell-binding between the authenticated data registry state and the ciphertext generation layer, precluding malicious data substitution by the seller. Finally, the framework cryptographically locks the generated ciphertext against a SHA256-hashed commitment of the ephemeral key exchange point, exposing it directly to the ILP conditional routing rails. We provide a full reference implementation utilizing `halo2-lib` over the BN254 field, demonstrating that Mandazi achieves highly succinct proving times and constant-time verification suitable for trustless, production-grade decentralized data marketplaces.

1 Preliminaries & Mathematical Foundations

The Mandazi framework coordinates state verification and stream privacy by embedding cryptographic primitives directly within the native scalar field of the BN254 elliptic curve, denoted as $\mathbb{F}_r$, where:

$$r = 2188824287183927522246405745257275088548364400416034343698204186575808495617$$

1.1 The Embedded Baby Jubjub Curve

To completely bypass costly non-native field math inside the arithmetic circuits, all asymmetric cryptographic operations are executed over the Baby Jubjub curve. Baby Jubjub is a twisted Edwards curve birationally equivalent to a Montgomery curve, defined natively over the base field $\mathbb{F}_r$ using standard parameter structures [3] by the algebraic relation:

$$ax^2 + y^2 = 1 + dx^2y^2$$

where the curve coefficients are configured as the elements:

$$a = 168700, \quad d = 168696$$

Let $G = (x_G, y_G)$ denote the fixed base generator point of prime order l , where:

$$l = 2736030358979909402780800718157159386076813972158567259200215660948447373041$$

A public key PK is generated via scalar multiplication of a localized private key scalar $sk \in \mathbb{F}_l$ such that $PK = sk \cdot G$.

1.2 Poseidon Duplex Sponge Permutation

For hash-based commitments, Merkle routing, and symmetric stream encryption, Mandazi utilizes the algebraic Poseidon permutation function optimized for circuit constraint scaling [2]. The internal state matrix $\mathbf{S}$ consists of t field elements representing the capacity c and rate R_{rate} lines:

$$\mathbf{S} = [s_0, s_1, \dots, s_{t-1}]^T \in \mathbb{F}_r^t$$

The state transition function executes $R_F = 8$ full rounds and $R_P = 57$ partial rounds. Each round applies a non-linear S-box substitution layer $f(x) = x^5$, an affine Matrix multiplication using an MDS matrix $\mathbf{M} \in \mathbb{F}_r^{t \times t}$, and round-constant additions $\mathbf{C}$:

$$\mathbf{S}_{j+1} = \mathbf{M} \cdot \mathbf{S}_j^5 + \mathbf{C}_j$$

In a Duplex Sponge configuration, data elements are absorbed sequentially into the state rate cells, and a pseudo-random keystream is squeezed out to drive the encryption matrix:

$$K_i = \text{Poseidon.squeeze}(\mathbf{S})$$

2 The Mandazi Circuit Architecture & Relations

The core technical contribution of the Mandazi circuit layout is the explicit containment of data validation and key derivation within a singular, unified PLONKish execution trace, compiling the full constraint system inside a PLONKish system matrix [4].

2.1 Monolithic Cell-Binding Semantics

To prevent an anonymous seller from executing a data-substitution attack (e.g., providing a valid Merkle path for an authentic registry leaf but encrypting dummy or corrupted data), Mandazi implements strict **in-circuit cell-binding**. In a PLONKish arithmetization grid consisting of advice columns **A**, fixed columns **F**, and selector columns **q**, a single, continuous witness allocation cell index w_{leaf} is instantiated. The circuit structure forces an absolute equality constraint by routing w_{leaf} concurrently as the leaf input to the Merkle lookup argument chips and the additive cipher gate: This structural configuration ensures that

Table 1: PLONK Execution Trace Matrix

Advice A	Advice B	Selector q	Enforced Constraint
w_{leaf}	π_{sibling}	q_{hash}	$\text{Poseidon}(w_{\text{leaf}}, \pi_i) = \text{Node}$ (Merkle Inclusion)
w_{leaf}	K_i	q_{add}	$w_{\text{leaf}} + K_i - \text{Cipher}_i = 0$ (Stream Encryption)

if the prover attempts to mutate the data payload to be encrypted, the copy constraint ($w_{\text{leaf}} \equiv M_i$) breaks the permutation argument of the global circuit, instantly rendering the ZK proof invalid.

2.2 Formal Gate Relations

The system layout compiles the full witness trace by simultaneously constraining five discrete mathematical identities:

1. **Ephemeral Key Commitment:** The circuit ingests a private ephemeral scalar $r \in \mathbb{F}_l$ and constrains its multiplication against the base generator point G to match the public input point C_1 :

$$C_1 = (x_{C_1}, y_{C_1}) = r \cdot G$$

2. **Diffie-Hellman Shared Secret Derivation:** Using the buyer's public key PK_{buyer} , the circuit computes the shared secret coordinate point S :

$$S = (x_S, y_S) = r \cdot PK_{\text{buyer}}$$

3. **Merkle Inclusion Proof:** The circuit ingests the private dataset leaf witness w_{leaf} along with its authentication path vector $\vec{\pi}$, asserting its membership against the public root R :

$$\mathcal{H}_n = \text{Poseidon}(\mathcal{H}_{n-1}, \pi_n) \implies \mathcal{H}_{\text{top}} - R = 0$$

4. **Duplex Stream Encryption:** The coordinates (x_S, y_S) initialize the internal Poseidon rate elements. The circuit enforces that the public ciphertext array $Cipher_i$ corresponds directly to the verified witness cell w_{leaf} :

$$K_i = \text{Poseidon}(x_S, y_S, i)$$

$$Cipher_i - w_{\text{leaf}} - K_i \equiv 0 \pmod{r}$$

5. **Interledger Point-Lock Integration:** To bridge the algebraic curve math to the out-of-circuit Rafiki payment engine, the circuit bit-decomposes the public coordinates of C_1 and asserts that its SHA-256 hash matches the public escrow routing condition H_{lock} :

$$\text{SHA256}(\text{Serialize}(x_{C_1}, y_{C_1})) == H_{\text{lock}}$$

2.3 Security Rationale: Why Each Constraint Exists

Each identity in the gate relation set functions as a cryptographic lock against a specific adversarial maneuver. Remove even one constraint, and the trustless marketplace collapses. The following breakdown maps every hoop to the exact security flaw it cures.

Hoop 1: Ephemeral Key Commitment ($C_1 = r \cdot G$). **Why jump through it:** The seller must commit to a specific, unique session coordinate before any data moves. **The attack it cures:** *Preventative identity theft and session replay*. Without binding r directly to a hard elliptic curve point (C_1) natively inside the trace, the seller could reuse old keys or switch out session parameters mid-computation, causing proof verification to desynchronize from the actual encryption stream.

Hoop 2: Diffie-Hellman Shared Secret Derivation ($S = r \cdot PK_{\text{buyer}}$). **Why jump through it:** The circuit derives a secure symmetric key that *only* the designated buyer can reconstruct out-of-circuit. **The attack it cures:** **The Interception / Eavesdropping Attack**. If payload encryption relied on a public variable readable from the ledger, routing intermediaries could steal the data. Diffie-Hellman lets intermediate nodes route payments blindly while the buyer retains exclusive decryption rights.

Hoop 3: Merkle Inclusion Proof ($\mathcal{H}_{\text{top}} - R = 0$). **Why jump through it:** The sold data must be legitimate and registered in an authoritative system (e.g., a valid KYC registry or CRB list). **The attack it cures:** **The “Snake Oil” or Garbage Data Attack**. An anonymous seller could encrypt fabricated data, generate a valid encryption proof, and steal the buyer’s payment. This hoop forces provenance verification against a trusted root R before a single byte is encrypted.

Hoop 4: Duplex Stream Encryption ($Cipher_i - w_{\text{leaf}} - K_i \equiv 0$). **Why jump through it:** The verified registry data cell is bound directly into the encryption gate. **The attack it cures:** **The Bait-and-Switch Attack**. This is the core of Mandazi. Without routing the *exact same* witness cell (w_{leaf}) through both the Merkle path and the cipher gate, a seller

could prove a valid KYC leaf while encrypting an entirely different payload. This constraint forces encryption of the *exact* asset that was proven.

Hoop 5: Interledger Point-Lock Integration ($\text{SHA256}(C_1) = H_{\text{lock}}$). **Why jump through it:** Algebraic curve coordinates are translated into a standard hash-link that existing web-payment infrastructure (Rafiki, Interledger conditional rails) can enforce natively. **The attack it cures: The Delivery-vs-Payment Failure.** Escrowed funds are locked to the data release. To withdraw via an Interledger connector or automated M-Pesa push, the seller must reveal C_1 to routing nodes. The moment C_1 appears on the wire, the buyer intercepts it, computes S , and unlocks the data stream.

*“We jump through these hoops to achieve **Atomic Trustlessness**. Hoops 1 and 2 lock down data privacy for the buyer; Hoop 3 locks down data authenticity from the registry; Hoop 4 enforces witness binding so the seller cannot switch the data out; and Hoop 5 ties the entire cryptographic engine directly into the Interledger financial payment rails for atomic clearance.”*

2.4 Execution and Atomic Settlement Flow

The transaction lifecycle progresses across five sequential phases, routing asset constraints natively over Open Payments environments via GNAP requirements [6] and Interledger engines [5]: During Phase 5, the seller triggers an automated M-Pesa Express STK Push or

Phase	Action	Cryptographic Payload
1	Handshake	Buyer transmits PK_{buyer} to Seller
2	Proving	Seller compiles circuit $\mathcal{R}$ and generates Proof π
3	Delivery	Seller sends $\{\pi, C_1, \text{Cipher}_i, H_{\text{lock}}\}$ to Buyer
4	Verification	Buyer validates π against R and locks funds to H_{lock} via Rafiki
5	Settlement	Seller claims funds via Rafiki by revealing $C_1 \implies$ Buyer decrypts stream

Figure 1: The Mandazi Protocol Execution Flow

Interledger contract settlement by revealing the preimage point C_1 to the network routing connectors to withdraw the escrowed funds. Because the Interledger plane only observes the hash-link relation $H_{\text{lock}} = \text{SHA256}(C_1)$, intermediate nodes gain zero advantage in interpreting the data stream. The buyer intercepts C_1 off the wire, executes a localized, out-of-circuit ECDH scalar multiplication using their private key sk_{buyer} to derive $S = sk_{\text{buyer}} \cdot C_1$, initializes the matching Poseidon sponge, and successfully decrypts the plaintext registry feed $M_i = \text{Cipher}_i - K_i$.

3 Conclusion & Future Horizon

Mandazi presents a novel paradigm for decentralized data commerce, successfully bridging the architectural chasm between Web-standard financial routing primitives operating on Ed25519 and high-performance zero-knowledge execution environments operating over the BN254 field. By establishing a strict, dual-curve lane-isolated protocol boundaries and enforcing monolithic, in-circuit atomic witness cell-binding, the framework entirely mitigates counterparty fraud and data-mutation vectors without imposing the prohibitive overhead of non-native field simulation. What Mandazi fundamentally unlocks is the tokenization of authenticated trust over existing open payment infrastructure. For the first time, highly sensitive identity layers—such as real-time Know-Your-Customer (KYC) statuses, Credit Reference Bureau (CRB) standing, or sovereign compliance state assertions—can be trustlessly streamlined and monetized on a per-query basis over standard Interledger (ILP) routing nodes and digital rails like Rafiki or automated mobile money API loops. By guaranteeing that data privacy is structurally tethered to atomic financial settlement via a Diffie-Hellman point-lock escrow, Mandazi obsoletes the need for centralized, vulnerable data honeypots. This architecture clears a concrete engineering path toward a highly scalable, sovereign, and privacy-preserving global data marketplace, laying the mathematical foundation for the next generation of trustless open-banking API protocols.

4 References

References

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